ECDLP index-calculus follow-on leads and experiment gates

Date: 2026-10-12. **Status:** source-bound search and experimental design. This round ran no new cryptographic measurement. It continues the conversation lead assessment and the N53 root-batch result. The N53 result is in open PR #1740, so its measured values are provisional until review. The private conversation export is neither reproduced nor used as a measurement source here.

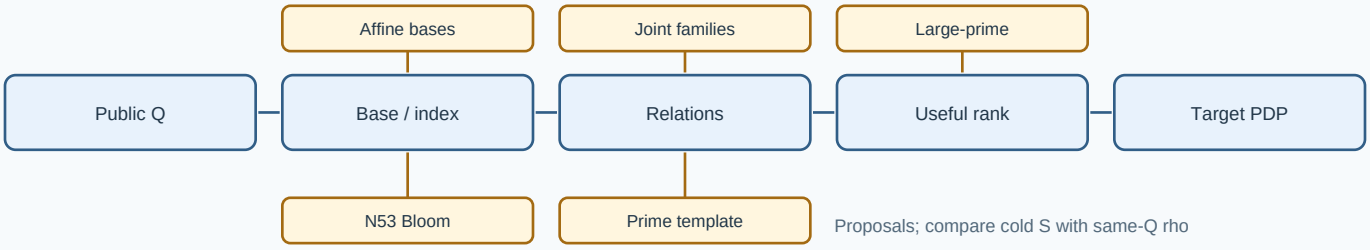
Boundary and present bottlenecks

The reference for each proposed complete solve is same-point, automorphism-aware rho with the same resource envelope. For a single target, charge base construction, relation collection, rank, sparse linear algebra, target decomposition, recovery, verification, failed attempts, and reference setup. The generic-group floor is expressed in `S = complete group-addition equivalents / sqrt(r)`; it must be recomputed for any changed base or orbit policy. A wall ratio, source-operation count, or prepared-online ratio is a diagnostic until the common unit is calibrated and the repository's replay and isolation gates pass.

The N53/K220 batch branch reports 220 novel rows from 220 accepted relations and median candidate/control cold 0.738 on nine pairs, yet candidate/strong-rho cold 52.586. Thus the immediate N53 question is where preparation and target PDP work remains after the root change. At N41, a historical coupled six-pair pipeline recovered 26 factor logs and one independent point, while its two point controls differed sharply in probe count (59,042 versus 730,852); its local timing does not estimate a distribution (`cryptanalysis:experiments/ecc2k130-ic-goal-20260929/n41_coupled_sixpair_full_ic_20261010/RESULT.md`). A separate local, unmerged `cryptanalysis` search found reciprocal rank 40 and full degree-two feature rank 821/821 for 3,544 column-compatible suffix sums. That exact finite-panel obstruction concerns **output-only** linear or quadratic equalities on that source; it does not rule out a condition coupling unresolved left and right sums. Source: `cryptanalysis:experiments/ecc2k130-ic-goal-20260929/n41_suffix_quadratic_span_20261012/RESULT.md` (local status, requiring native replay before adoption).

The lead diagram identifies the stage tested by each route. Its editable source is lead-gates.mmd.

Where each proposal enters the charged pipeline



Active follow-ons from the reviewed leads

1. N41 joint family elimination, then N131 projection. The working curve is `EC1N41Ckb1h023d9a38bbf7`, `GF(2)[t]/(t^41+t^3+1)`, $y^2+xy=x^3+1$, subgroup order 549,756,390,943. The N41 six-pair result supplies a separate complete relation/rank control; the later finite-family join is exact only for a restricted rotation-zero shared-seed family and missed its frozen public point. The next algorithm must keep at least two unresolved pair outputs on each side and reject *whole coefficient families before opening source masks*. Port the relevant path to native Rust first, since the historical `cryptanalysis` executors and auditors use Python. Freeze the 12 positive **eight-column** rows of the existing public-point fiber, ordinary public points, source masks, and full family census before evaluating a predicate. Require every positive row to survive, zero false rejects by independent group replay, an exhaustive family accounting, and lower complete charged work than the exact direct suffix join on the same source. Measure natural accepted relations and novel rank on at least three fresh public points. Only a passing N41 gate permits a source-bound N131 projection under one worker, 600 seconds, and 4 GiB; a projected pass is still separate from a completed N131 query. A timeout remains censored. The precise protocol records the frozen admission sequence.

2. N53 index membership after root batching. `KIC_ROOT_BLOOM` is already an exact optional root-table filter in the N53 native example; its tests establish no false negatives for inserted keys, while no same-target cost panel has been run with the new batch path. Compare Bloom off/on with batching enabled, a fixed K220 base, three new public `Q` and three rank seeds per `Q`, then matched strong rho. Check identical base/index contents, rank trace, target scalar, and replay receipts; count exact lookups, filter negatives, false positives, construction work and bytes. Advance this engineering path only if all cells are correct, candidate/control median complete cold is at most 0.95, online at most 0.90, and peak RSS at most 1.10 of control under the declared envelope. Those thresholds are screening choices, not evidence. A source-level operation audit is needed before common-S calibration, and an isolated host is needed before a CPU claim. This is distinct from the mathematically harder N41 relation-supply gate. The N53 screen protocol gives the exact follow-on admission sequence.

Additional leads found in primary research

3. Slot-specific affine factor bases. A characteristic-two summation-polynomial study proposes distinct translated subspaces for separate summand slots, reporting a relation-probability benefit in its own setting (Galbraith and Gebregiyorgis, ePrint 2014/806). The existing rotated-window experiments already use nonidentical support, so the new question is narrower: does an *affine offset selected for disjoint folded columns* improve verified relation yield per charged work at a fixed total of useful orbit columns? Pre-register offset choices using training points, then hold out public N37/N41/N53 points. Keep total column count, relation length, and actual base size explicit, charge every extra index and factor log, and measure verified relations, novel-rank fraction, target PDP, complete cold cost, and memory. A twofold gain in **novel rows per charged query** without a cold-cost regression is a stage gate; complete S and matched rho decide any method claim. The offset selection must not inspect held-out targets.

4. One- or two-large-prime partial-relation graph. The double-large-prime mechanism is established for small-genus hyperelliptic index calculus (Gaudry et al., ePrint 2004/153); its benefit does not transfer automatically to these elliptic factor bases. Here it is a *new proposal*: accept verified relations with one or two designated external orbit columns, then combine graph cycles to eliminate them before modular-rank insertion. First establish a native exact partial producer on N41 and N53 without changing the complete base. Freeze the external support and graph memory budget. Independently replay each partial relation and every resulting cycle row; reject trivial collisions and count dependent cycles. Compare novel full rows per total source operation, graph construction/retention cost, matrix density, and cold single-target cost with ordinary relations on the same public points. Stop this route if cycle formation produces fewer novel rows per charged work, or memory exceeds the declared cap. A wider accepted support changes the counting floor and must be reported as a policy comparison.

5. Reusable elimination templates for prime-field systems. A prime-field paper reduces repeated Gröbner-basis calculations to one reusable calculation in its formulation (Amadori, Pintore and Sala, ePrint 2017/609). Test the *scope of reuse*, not a claimed speedup, on one exact prime-field curve/subgroup. Freeze a target-independent ideal/template, verify symbolic specialization against independently computed summation-polynomial evaluations, then run ordinary target points with all specialization, solve, and failure costs. If the reusable computation incorporates target data, charge it per target. Require a verified relation and rank contribution before timing a full solve. This track does not imply a binary Koblitz improvement or an executable cover/isogeny.

Decision table

Route	Immediate test and boundary	Move forward when
Joint N41 family elimination	Exact direct suffix join and frozen positive rows; N131 600 s/4 GiB only after native N41 gate	All known rows retained, ordinary relations and novel rank measured, whole-family rejection occurs before expansion, lower total N41 work
N53 Bloom after batching	Same-Q/seed batch control and strong rho; fixed K220	Exact replay, cold ratio ≤0.95, online ratio ≤0.90, RSS ratio ≤1.10; then common-S and isolation
Affine slot bases	Fixed useful columns and complete cold/rho	≥2 novel rows per charged query relative to control with no cold regression, then held-out full solve
Large-prime graph	Ordinary full relations on same support/points	More novel full rows per total work after cycles, bounded memory and matrix fill
Prime-field template	One frozen exact curve and matched target solver	Verified target-independent reuse and ordinary relation/rank benefit after all setup is charged

The immediate execution order is the native N41 joint gate and the N53 membership panel. The three additional routes are proposals with no new experiment in this PR. Do not promote an N41 component, an N53 stage ratio, or a reusable-template count to a complete-method claim. A verified same-target scalar, replayed factor logs and rank, complete operation accounting, common calibrated S, and matched strong rho remain the claim gate.

Requirement and graph status

User request	Status	Evidence or remaining work
Pursue assessed leads	Partial	Reconciled N41/N53 bottlenecks and specified executable gates; no new run in this search round.
Find other leads	Complete for this search round	Three source-linked proposals above, each with a transfer obligation and a falsifier.
Devise experiments and criteria	Complete for this search round	The decision table and N41 admission protocol specify controls, costs and stop conditions.

Checked `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json`, and the prior N53 graphs. They are unchanged: this search produces no new verified measurement or correction to a canonical series. The attached PDF is a rendering of this report and diagram, not a separate evidence source.